

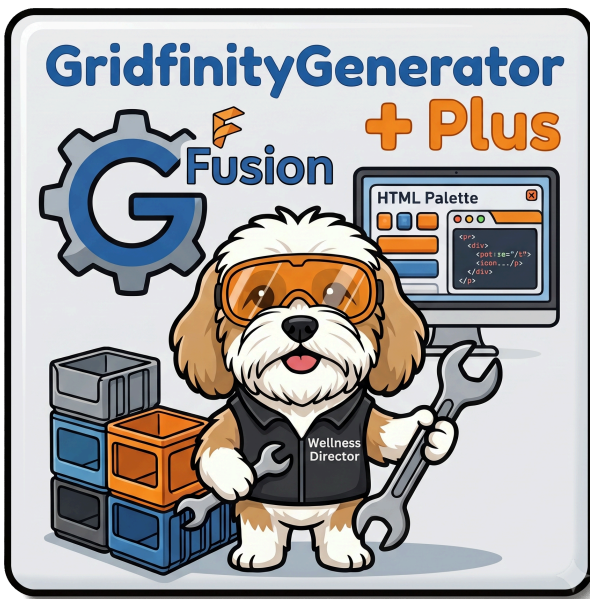
GridfinityGeneratorPlus

Version: 1.1.0.0

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A Fusion add-in designed to make generating, previewing, and revisiting Gridfinity parts fast, intuitive, and future-proof with creation settings that aren't lost to time.



Introduction: The "Why" and "What"

The original GridfinityGenerator add-in is a great parametric bin/baseplate generator, but it works through two separate command dialogs that don't share settings, offer no preview before committing, and forget everything about how a part was built the moment you click OK — so revisiting a design months later, or handing the file to someone else, means starting from scratch.

GridfinityGeneratorPlus keeps the original's proven geometry engine and rebuilds the interface and workflow around it.

GridfinityGeneratorPlus is a Fusion add-in designed to make generating, previewing, and revisiting Gridfinity parts fast and durable.

- **Feature 1: One persistent palette, not two dialogs.** Bin, Baseplate, Grid Optimizer, and Theme all live in a single dockable palette with shared "Common settings," live Update Preview /

Generate / Clear Preview, and named saved configurations per tab.

- **Feature 2: Settings travel with the file.** Every generated part remembers the exact parameters that built it, stored directly on the component itself. Reopen a design months later, or share the .f3d with someone else running this add-in, and click **Edit Active Component** to pick up right where you left off.
- **Feature 3: Grid Optimizer.** Building a custom (non-42mm) grid to fit a specific drawer or cabinet? Enter your real measurements and let the optimizer recommend a pitch that minimizes wasted space, instead of guessing.
- **Feature 4: Stackable baseplates.** A fourth baseplate type, made symmetric top-to-bottom so it can be printed and physically stacked with no overhang, with configurable interface layers between plates.
- **Feature 5: Theme tab.** Customize the palette's appearance — built-in themes, automatic OS dark-mode following, and support for importing your own custom themes.

What's New in v1.1.0

- **Grouped Undo:** Update Preview and Generate each collapse to a single Undo step, instead of leaving several separate entries on Fusion's Undo stack.
- **Undo/Redo-aware preview tracking:** the Clear Preview button now stays in sync as you step through Fusion's Undo/Redo — including recognizing an earlier preview reappearing several steps back, not just the most recent one.
- **Cross-tab preview warning:** switching to the other tab and clicking Update Preview or Generate now warns you first if it would discard an unfinalized preview you built on the tab you're leaving.
- **Redesigned Configurations manager:** Load, Delete, and Save As now each open a focused dialog instead of sharing an inline dropdown, with a clear "Active config" status line — Update Current is now the only button that acts on the current selection directly.
- **Custom themed dialogs:** every native browser confirm/prompt popup in the palette has been replaced with a themed HTML modal that matches whichever theme you have selected.
- **Theme tab: font controls, export, and cleanup.** Choose a font family and base font size on top of any theme, export the current look (including your font tweaks, which now correctly round-trip on re-import) as a shareable .theme.json via a native Save dialog defaulted to resources/themes/ , and remove imported themes you no longer want.
- **Linked Factory Reset:** resetting Bin or Baseplate now also resets the shared Common settings (base unit, clearance, magnet size), and vice versa — previously each Factory Reset button only reset its own section, so a loaded custom-grid preset could leave Common looking un-reset.

- **Sortable Grid Optimizer dimensions table:** click a column heading (Description/Width/Depth) to sort your dimension list, click again to reverse — the order you leave it in is what gets saved.
- **Visual polish:** a more distinct tab bar in dark themes, and hint/error text that now scales with the base font size setting.

For the full development history and the reasoning behind each design decision, see [docs/Dev_Notes.md](#).

Installation

Tip: For a stable, versioned copy, download the latest packaged release from the **Releases** link in the right-hand sidebar of this repo's GitHub page instead of using the green **Code** button above — the Code button always pulls the current `main` branch, which may include in-progress changes. Or, simply click this link to the [Releases page](#)

Manual Installation Options

This add-in requires a quick manual installation. You can choose to install it in Fusion's default directory or a custom folder of your choice.

Option 1: Install in the Default Fusion Directory

1. **Download:** Download the source code as a ZIP file and extract the `GridfinityGeneratorPlus-main` folder. Rename the folder to `GridfinityGeneratorPlus` (remove the `-main` suffix) — Fusion requires the folder name to match the add-in name exactly, so it won't run correctly if you skip this step.
Download the zip file using the green `code` button above or simply click this link: [GridfinityGeneratorPlus Main Branch](#)
2. **Move the Folder:** Move the entire `GridfinityGeneratorPlus` folder into your native Fusion Add-Ins directory:
 - **Windows:** `%appdata%\Autodesk\Autodesk Fusion 360\API\Addins`
 - **Mac:** `~/Library/Application Support/Autodesk/Autodesk Fusion 360/API/Addins`
3. **Open Fusion:** Press `Shift + S` to open the **Scripts and Add-Ins** dialog.
4. **Run the Add-in:** Make sure the **Add-ins** filter checkbox is checked. You should see **GridfinityGeneratorPlus** in the list of add-ins. You may want to check the 'Run on startup' option so it automatically runs when Fusion starts. Click the **Run** icon to execute the add-in.

Option 2: Install in a Custom Directory

1. **Download:** Download the source code as a ZIP file and extract the `GridfinityGeneratorPlus` folder. Rename the folder to `GridfinityGeneratorPlus`.
2. **Organize:** Create a dedicated folder on your computer for your Fusion tools (e.g., `Documents\Fusion_Tools`) and move the `GridfinityGeneratorPlus` folder inside it (remove the `-main` suffix) — Fusion requires the folder name to match the add-in name exactly, so it won't run correctly if you skip this step.
3. **Open Fusion:** Press `Shift + S` to open the **Scripts and Add-Ins** dialog.
4. **Add the Add-in:** Click the grey "+" icon next to the search box at the top of the dialog and select **Script or add-in from device**.
5. **Locate:** Navigate to your custom folder, select the `GridfinityGeneratorPlus` folder, and click **Select Folder**.
6. **Run the Add-in:** Make sure the **Add-ins** filter checkbox is checked. You should see **GridfinityGeneratorPlus** in the list of add-ins. You may want to check the 'Run on startup' option so it automatically runs when Fusion starts. Click the **Run** icon to execute the add-in.

Using GridfinityGeneratorPlus

The Gridfinity Generator Palette

In the **DESIGN** workspace, find the **SOLID** toolbar panel's **CREATE** dropdown and select **"GridfinityGeneratorPlus"** to open the palette. It stays docked and open across your session — set your Common settings once, then switch between tabs as needed.

- **Common settings panel:** base grid unit, XY clearance, and magnet cutout size, shared by both Bin and Baseplate so they always stay compatible with each other.
- **Update Preview / Generate / Clear Preview:** preview a part in the viewport before committing to it, make it permanent when you're happy, or discard the preview entirely.
- **Configurations:** save, load, and delete named presets per tab (Bin, Baseplate, Grid Optimizer), so a frequently-used setup is always one click away.

Edit Active Component

Activate a previously generated Gridfinity part in the Fusion browser by hovering over the component name and clicking the radio button that appears to the right of the name, then click **Edit Active Component** at the top of the palette — it reloads the exact settings used to build that part.

- **Tip:** this requires the design to be in **Hybrid** design mode (Document Settings → Design → Hybrid). If you click **Generate** in a Part or Assembly design, the palette will prompt you to switch modes — your entered values aren't lost, so just switch and click Generate again.

Tech Stack

For the fellow coders and makers out there, here is how GridfinityGeneratorPlus was built:

- **Language:** Python (Fusion API)
- **Interface:** HTML/CSS/JavaScript (Palette)
- **Data Storage:** Each generated part's settings are stored as a Fusion component attribute directly tied to the component, so they travel with the .f3d file itself. Add-in preferences (window geometry, active theme), saved configurations, and imported themes are stored locally as JSON alongside the add-in.

Acknowledgements & Credits

- **Original Work:** GridfinityGeneratorPlus is a fork of [GridfinityGenerator](#) by **Lev Mishin**. The core parametric geometry — the base/foot profile, bin body construction, and baseplate generation — originates from his original work, without which this fork wouldn't exist.
- **Developer:** Ed Johnson ([Making With An EdJ](#))
- **AI Assistance:** Developed with coding assistance from Claude (Anthropic).
- **Lucy (The Cavachon Puppy):**
Chief Wellness Officer & Director of Mandatory Breaks
 - Thank you for ensuring I maintained healthy circulation by interrupting my deep coding sessions with urgent requests for play.
- **License:** Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

Support the Maker (and Lucy!)

I develop these tools to improve my own workflows and love sharing them with the community. If you find GridfinityGeneratorPlus useful and want to say thanks, feel free to [buy Lucy a dog treat on Ko-fi!](#)

Happy Making!

— EdJ